

Microeconomics for Finance

Decision Making under Uncertainty

Jérôme Dugast



Choosing Between Uncertain Payments

	Payments, Good state	Payments, Bad state
Asset 1	1500	1000
Asset 2	2000	500
Asset 3	2000	1000

Choosing Between Uncertain Payments

	Payments, Good state	Payments, Bad state
Asset 1	1500	1000
Asset 2	2000	500
Asset 3	2000	1000

Asset 3 payments **state by state dominates** those of assets 1 and 2.

⇒ Obviously, 3 is preferred

Asset 1 vs. Asset 2 is not obvious : more in state B but less in state G.

Choosing Between Uncertain Payments

	Payments, Good state	Payments, Bad state
Asset 1	1500	1000
Asset 2	2000	500
Asset 3	2000	1000

Asset 3 payments **state by state dominates** those of assets 1 and 2.

⇒ Obviously, 3 is preferred

Asset 1 vs. Asset 2 is not obvious : more in state B but less in state G.

Similar problem. Which basket the consumer does prefer : 2 apples and 3 bananas or 3 apples and 2 bananas ?

⇒ We must quantify the **agents' preferences** with a **utility function**.

Properties of the Utility Function

We want to determine a utility function, U , “reasonable”, which generates a **relation of order** (i.e, a preference) over the set of risky payments (lotteries).

Back to our example, the utility function should depend positively on the payment realization in each state of the world.

If $\tilde{a}$ is a risky payments with realizations a^G, a^B , then

$$U(\tilde{a}) = U(a^G, a^B) \text{ with } \frac{\partial U}{\partial a^G} > 0, \frac{\partial U}{\partial a^B} > 0$$

Properties of the Utility Function

How to take into account the probabilities of each state of the world?

	Payments, Good state	Payments, Bad state
Asset 1	1500	1000
Asset 2	2000	500

- If $\pi_G = 0.99$ and $\pi_B = 0.01$, we should prefer asset 2.

Properties of the Utility Function

How to take into account the probabilities of each state of the world?

	Payments, Good state	Payments, Bad state
Asset 1	1500	1000
Asset 2	2000	500

- If $\pi_G = 0.99$ and $\pi_B = 0.01$, we should prefer asset 2.
 - Denote $u(x)$ the utility obtained from a **certain payment** x
- a good “candidate” is an **expected utility function**

$$U(\tilde{a}) = \pi_G u(a^G) + \pi_B u(a^B) = E[u(\tilde{a})]$$

Properties of the Utility Function

	Payments, Good state	Payments, Bad state
Asset 1	1500	1000
Asset 2	2000	500

- If $\pi_G = \pi_B = 0.5$, both assets have same mean 1250. Which one do you prefer?

Properties of the Utility Function

	Payments, Good state	Payments, Bad state
Asset 1	1500	1000
Asset 2	2000	500

- If $\pi_G = \pi_B = 0.5$, both assets have same mean 1250. Which one do you prefer?
- If you are **risk averse**, $E[u(\tilde{a}_1)] > E[u(\tilde{a}_2)]$.

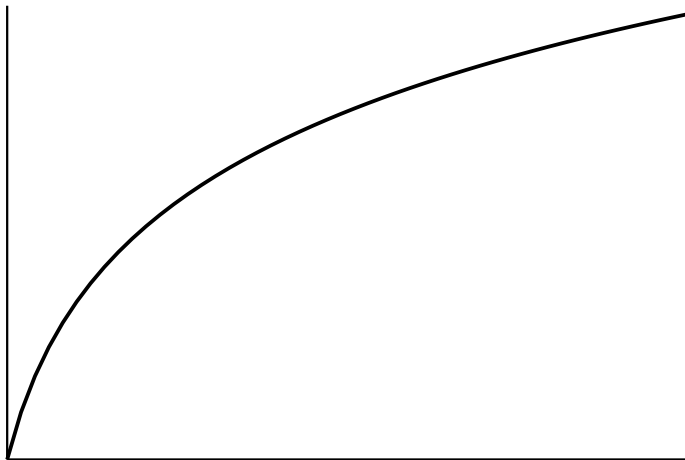
→ If we still consider the “candidate”

$$E[u(\tilde{a})] = \pi_G u(a^G) + \pi_B u(a^B),$$

Which condition on u ?

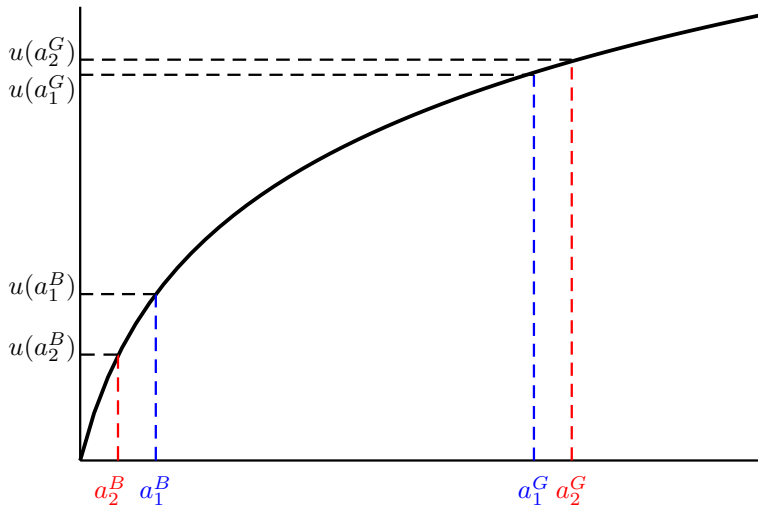
Concavity

u concave



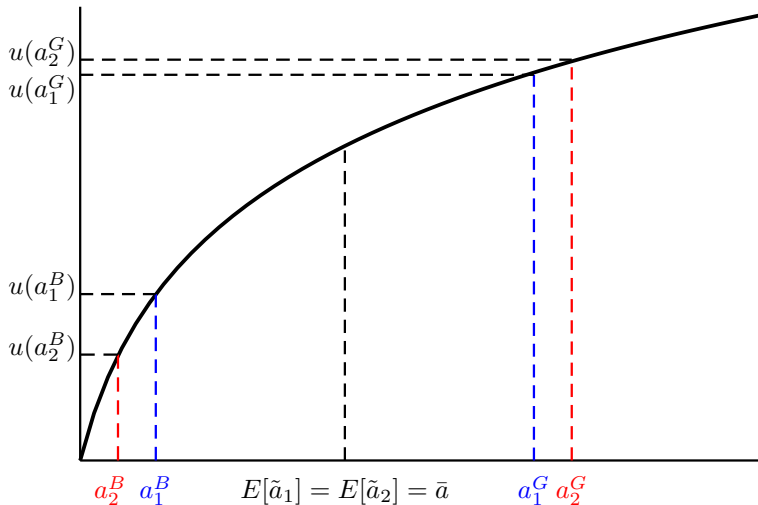
Concavity

u concave



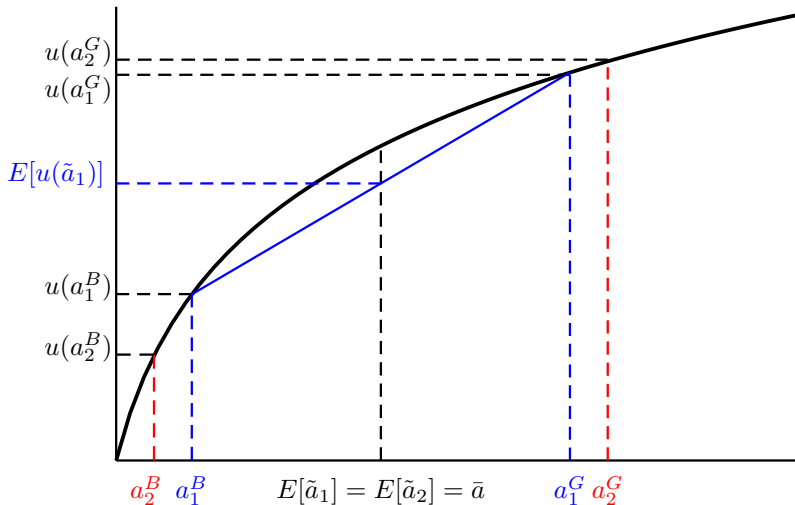
Concavity

u concave

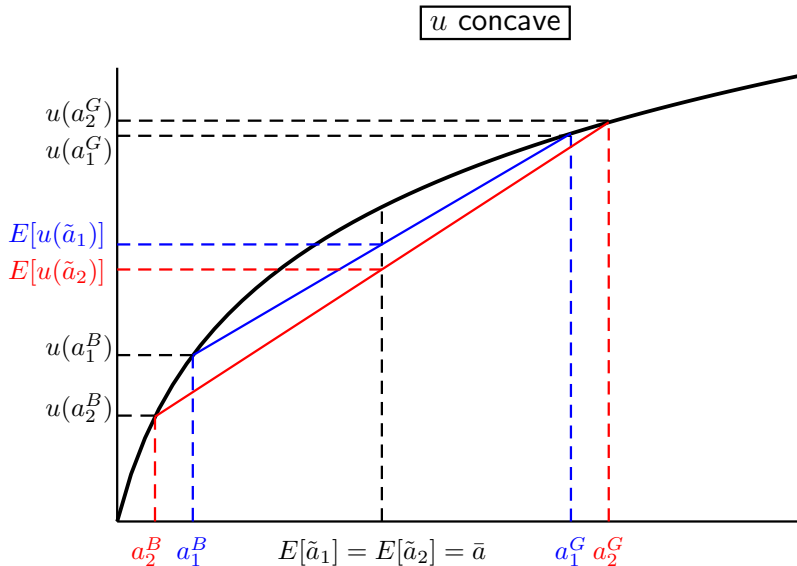


Concavity

u concave

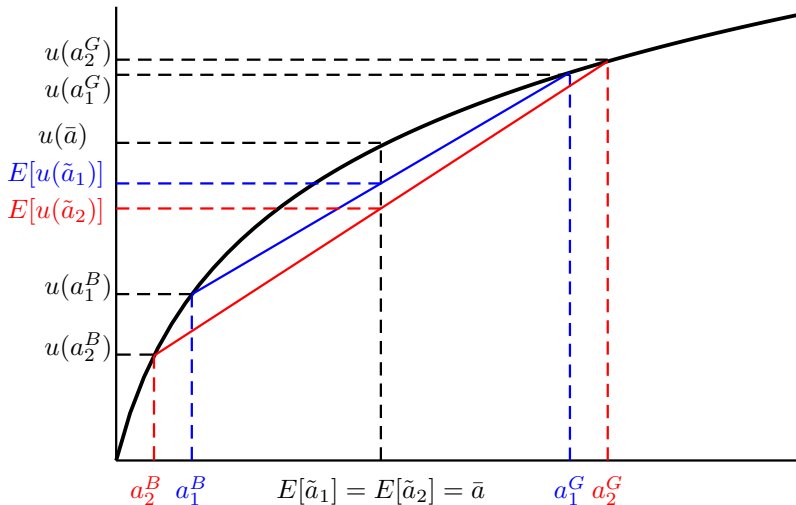


Concavity



Concavity

u concave



Expected Utility

John **von Neumann** and Oskar **Morgenstern** have determined conditions under which agents' preferences for risky payments (lotteries) can be described by an **expected utility function** such that

$$E[u(\tilde{a})] = \sum_{n=1}^N \pi_n u(a_n)$$

with u **concave and increasing**, $u' > 0$ and $u'' < 0$.

More generally, if a risky payment $\tilde{a}$ has a distribution $d\Phi$,

$$E[u(\tilde{a})] = \int_{\mathbb{R}} u(a) d\Phi(a)$$

Bibliography. The axiomatic approach behind the theory of the expected utility function are developed by von Neumann and Morgenstern in their book *Theory of Games and Economic Behavior* (1947).

Risk Aversion

Under these conditions, agents are risk averse. In particular, with **Jensen inequality** for a concave function, we have

$$E[u(\tilde{a})] \leq u(E[\tilde{a}]).$$

Agents would prefer to obtain the mean of payment with certainty rather than the risky payment.

How to quantify risk aversion ?

The level of concavity, measured by the second derivative, u'' , **is not** a good measure. Indeed, if

$$v = \alpha u + \beta, \text{ with } \alpha > 1$$

v generates the same relation of order among risky payments than u does. Yet $v'' < u'' < 0$.

⇒ A measure of risk aversion must be invariant with any affine transformation of the utility function.

Measuring Risk Aversion

There are two classic measures of risk aversion

$$R_A(y) = -\frac{u''(y)}{u'(y)} = \text{coefficient of absolute risk aversion}$$

and

$$R_R(y) = -\frac{yu''(y)}{u'(y)} = \text{coefficient of relative risk aversion}$$

where y is the payment (the income) of the agent.

Both are positive : $R_A(y) > 0, R_R(y) > 0$.

Interpreting Risk Aversion

Consider a risk averse agent who receives a risky payment $y + \tilde{\varepsilon}$, with mean y and variance $Var[\tilde{\varepsilon}] = \sigma^2$.

The **certainty equivalent** of this payment is $y - \Delta$ such that

$$u(y - \Delta) = E[u(y + \varepsilon)]$$

Δ is a **risk premium** : How much the agent is willing to reduce his mean payment to get rid of uncertainty.

Interpreting Risk Aversion

Assuming that Δ and ε are small, $\Delta \ll 1$ and $\sigma^2 \ll 1$, give an approximation of Δ .

$$u(y - \Delta) \approx u(y) - \Delta u'(y)$$

$$E[u(y + \varepsilon)] \approx E[u(y)] + \underbrace{E[\varepsilon u'(y)]}_{=u'(y)E[\varepsilon]=0} + \frac{1}{2}E[\varepsilon^2 u''(y)] = u(y) + \frac{\sigma^2}{2}u''(y)$$

$$\Rightarrow \Delta = -\frac{u''(y)}{u'(y)} \frac{\sigma^2}{2} = R_A(y) \frac{\sigma^2}{2}$$

Interpreting Risk Aversion

When the risky payment is $y + y\tilde{\varepsilon}$, define δ as

$$u(y - \delta y) = E[u(y + y\varepsilon)].$$

Similarly, one can give a first order approximation of δ under the assumption $\delta \ll 1$ and $\sigma^2 \ll 1$ (cf Problem Set).

$$u(y - \delta y) \approx u(y) - \delta y u'(y)$$

$$E[u(y + y\varepsilon)] \approx u(y) + \frac{y^2 \sigma^2}{2} u''(y)$$

$$\Rightarrow \delta = -\frac{y u''(y)}{u'(y)} \frac{\sigma^2}{2} = R_R(y) \frac{\sigma^2}{2}$$

Whether uncertainty is absolute or relative (w.r.t y), the certainty equivalent depends on the coefficient of absolute or relative risk aversion.

Usual Utility Functions

- CARA (“Constant Absolute Risk Aversion”)

$$u(y) = -\frac{1}{\nu}e^{-\nu y}$$

- CRRA (“Constant Relative Risk Aversion”)

$$u(y) = \frac{y^{1-\gamma}}{1-\gamma}$$

and, in addition, $u(y) = \ln(y)$ for $\gamma = 1$.

- What are $R_A(y)$ in the CARA case, and $R_R(y)$ in the CRRA case?

Risk Neutrality

When u is affine, the agent is risk neutral :

$$u(y) = \alpha y + \beta$$

He only cares about his expected revenue.

Specifically,

$$R_A(y) = R_R(y) = 0 \quad \forall y$$

Risk Aversion and Portfolio Choice

Consider an agent with an expected utility function à la von Neumann-Morgenstern, who invests at $t = 0$ and, at $t = 1$, receives the pay-off of its investment and “consume” it (get the utility out of the payment).

The agent can invest his initial wealth y_0 in the following way :

- invests an amount w in a risky asset (the S&P 500 for instance) with random net return $\tilde{r}$,
- and the rest in the risk free asset with (certain) net return, r_f .

His final wealth $\tilde{y}_1$ at $t = 1$:

$$\begin{aligned}\tilde{y}_1 &= (y_0 - w) \times (1 + r_f) + w \times (1 + \tilde{r}) \\ &= y_0 \times (1 + r_f) + w \times (\tilde{r} - r_f)\end{aligned}$$

Risk Aversion and Portfolio Choice

Objective function of the agent :

$$\max_w U(w) = \max_w E[u(\tilde{y}_1)] = \max_w E[u(y_0(1 + r_f) + w(\tilde{r} - r_f))]$$

Unconstrained optimization of a concave function,

$$u'' < 0 \Rightarrow U''(w) = E[(\tilde{r} - r_f)^2 u''(y_0(1 + r_f) + w(\tilde{r} - r_f))] < 0$$

The optimal amount w^* is given by the first order condition :

$$U'(w) = 0$$

Risk Aversion and Portfolio Choice

First order condition :

$$E[(\tilde{r} - r_f)u'(y_0(1 + r_f) + w^*(\tilde{r} - r_f))] = 0$$

Remark : Unconstrained optimization implies that short-selling ($w < 0$) or leverage ($w > y_0$) are possible.

Proposition 1. When u is increasing and concave, $u' > 0$, $u'' < 0$, then,

$$w^* > 0 \Leftrightarrow E[\tilde{r}] > r_f$$

$$w^* = 0 \Leftrightarrow E[\tilde{r}] = r_f$$

$$w^* < 0 \Leftrightarrow E[\tilde{r}] < r_f$$

Implication. Whenever the expected return is larger than the risk free rate, it is always optimal to invest in (at least a little).

Risk Aversion and Portfolio Choice

Proof. Define $F(w)$ as

$$F(w) = E[(\tilde{r} - r_f)u'(y_0(1 + r_f) + w(\tilde{r} - r_f))]$$

w^* is such that $F(w^*) = 0$

Derivative of F :

$$F'(w) = E[(\tilde{r} - r_f)^2 u''(y_0(1 + r_f) + w(\tilde{r} - r_f))] < 0$$

Value of F in 0 :

$$F(0) = E[(\tilde{r} - r_f)u'(y_0(1 + r_f))] = \underbrace{u'(y_0(1 + r_f))}_{>0} \times (E[\tilde{r}] - r_f)$$

Risk Aversion and Portfolio Choice

If the investor is risk neutral :

$$u(y) = \alpha y + \beta$$

$$\max_w E[u(\tilde{y}_1)] \Leftrightarrow \max_w y_0(1 + r_f) + w(E[\tilde{r}] - r_f).$$

Then,

$$E[\tilde{r}] > r_f \Rightarrow w^* = +\infty$$

$$E[\tilde{r}] < r_f \Rightarrow w^* = -\infty$$

Risk Aversion and Portfolio Choice

Same problem, different formulation : The agent buys z units of risky assets, at price p , with pay-off at $t = 1$ equal to $\tilde{a}$.

$$\begin{aligned}\tilde{y}_1 &= (y_0 - p.z) \times (1 + r_f) + z \times \tilde{a} \\ &= y_0 \times (1 + r_f) + z \times (\tilde{a} - (1 + r_f)p)\end{aligned}$$

First order condition :

$$E[(\tilde{a} - (1 + r_f)p)u'(y_0(1 + r_f) + z^*(\tilde{a} - (1 + r_f)p))] = 0$$

$$z^* > 0 \Leftrightarrow \frac{E[\tilde{a}]}{1 + r_f} > p$$

Consistent with Proposition 1 because :

$$1 + E[\tilde{r}] = \frac{E[\tilde{a}]}{p}$$

Risk Aversion, Portfolio and Initial Wealth

Let $w^*(y_0)$ the solution of

$$\max_w E[u(y_0(1 + r_f) + w(\tilde{r} - r_f))], \text{ when } E[\tilde{r}] > r_f$$

which implies $w^* > 0$.

Proposition 2. If u is such that

$$R'_A(y) < 0 \Rightarrow \frac{dw^*}{dy_0} > 0$$

$$R'_A(y) = 0 \Rightarrow \frac{dw^*}{dy_0} = 0$$

$$R'_A(y) > 0 \Rightarrow \frac{dw^*}{dy_0} < 0$$

Risk Aversion, Portfolio and Initial Wealth

Proof. Assume that $R'_A(y) < 0$. By taking the derivative of the first order condition with respect to y_0 we get :

$$E \left[(\tilde{r} - r_f) \left(1 + r_f + \frac{dw^*}{dy_0} (\tilde{r} - r_f) \right) u''(\tilde{y}_1) \right] = 0$$

$$\begin{aligned} \frac{dw^*}{dy_0} \overbrace{E [(\tilde{r} - r_f)^2 u''(\tilde{y}_1)]}^{<0} &= (1 + r_f) E [(\tilde{r} - r_f)(-u''(\tilde{y}_1))] \\ &= (1 + r_f) \underbrace{E [(\tilde{r} - r_f) u'(\tilde{y}_1) R_A(\tilde{y}_1)]}_{=H} \end{aligned}$$

$$\begin{aligned} H &= E [(\tilde{r} - r_f) u'(\tilde{y}_1) R_A(\tilde{y}_1) | \tilde{r} < r_f] P[\tilde{r} < r_f] \\ &\quad + E [(\tilde{r} - r_f) u'(\tilde{y}_1) R_A(\tilde{y}_1) | \tilde{r} > r_f] P[\tilde{r} > r_f] \\ &\leq E [(\tilde{r} - r_f) u'(\tilde{y}_1) R_A(y_0(1 + r_f)) | \tilde{r} < r_f] P[\tilde{r} < r_f] \\ &\quad + E [(\tilde{r} - r_f) u'(\tilde{y}_1) R_A(y_0(1 + r_f)) | \tilde{r} > r_f] P[\tilde{r} > r_f] \\ &\leq R_A(y_0(1 + r_f)) E [(\tilde{r} - r_f) u'(\tilde{y}_1)] = 0 \end{aligned}$$

Risk Aversion, Portfolio and Initial Wealth

Let $\eta(y_0)$ the elasticity of $w^*(y_0)$:

$$\eta(y_0) = \frac{d \ln(w^*(y_0))}{d \ln(y_0)}$$

Proposition 3. If u is such that

$$R'_R(y) < 0 \Rightarrow \eta(y_0) > 1$$

$$R'_R(y) = 0 \Rightarrow \eta(y_0) = 1$$

$$R'_R(y) > 0 \Rightarrow \eta(y_0) < 1$$

Risk Aversion, Portfolio and Initial Wealth

Proof. Let $w^* = e^{\theta^*}$ and $y_0 = e^{\gamma_0}$,

$$\eta = \frac{d\theta^*}{d\gamma_0}.$$

By taking the derivative of the first order condition with respect to γ_0 we get :

$$E \left[(\tilde{r} - r_f) \left((1 + r_f)y_0 + \frac{d\theta^*}{d\gamma_0} w^* (\tilde{r} - r_f) \right) u''(\tilde{y}_1) \right] = 0$$

$$\begin{aligned} \left(\frac{d\theta^*}{d\gamma_0} - 1 \right) w^* \overbrace{E [(\tilde{r} - r_f)^2 u''(\tilde{y}_1)]}^{<0} &= E [(\tilde{r} - r_f) \tilde{y}_1 (-u''(\tilde{y}_1))] \\ &= E [(\tilde{r} - r_f) u'(\tilde{y}_1) R_R(\tilde{y}_1)] \end{aligned}$$